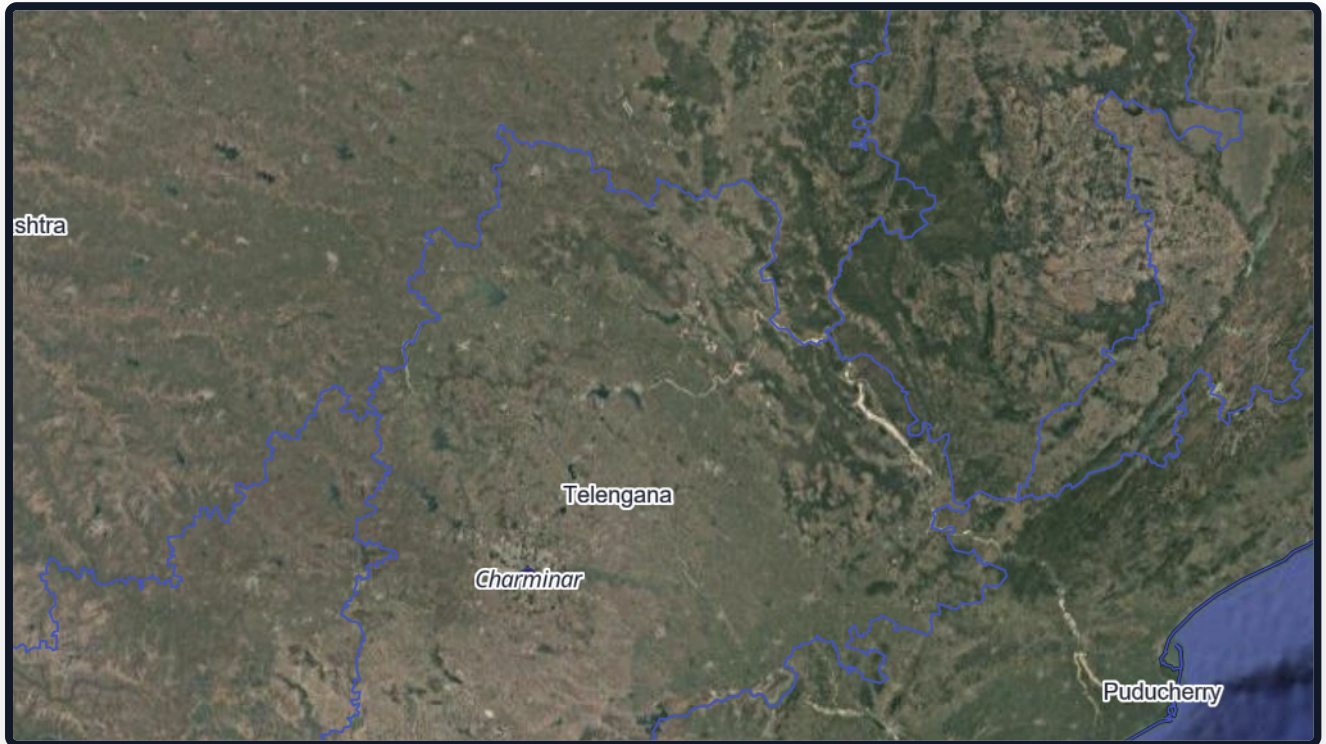


Famous Places of India

Put landmarks on the map using GeoLibre



An example of a famous place on a map. Yours will look different!

What you'll use

- **GeoLibre**: the same free, web-based GIS tool from Exercise 1
- **famous_places.csv**: a small table of 16 famous places, with their latitude and longitude
- **india_boundary.geojson**: the outline of India, from the Exercise 1 data folder

Every Place Has an Address on Earth

Your home has a street address. But every spot on Earth also has a special address made of two numbers, called **latitude** and **longitude**. With those two numbers, a GIS can put a dot in exactly the right place on a map.

Today you'll take a simple table of famous places in India, like the Taj Mahal, the Golden Temple, and Kaziranga National Park, and turn it into a map. Then you'll explore the places and measure how far apart they are.

GIS WORD OF THE DAY

Latitude and longitude: *latitude* tells you how far north or south of the equator a place is. *Longitude* tells you how far east or west it is. Together they are called **coordinates**.

FUN FACT

India sits entirely north of the equator, so every place in India has a positive latitude. The southern tip of the mainland, Kanyakumari, is at about 8 degrees north.

Here's the plan for today, in eight steps:

1. Open GeoLibre and add the India outline
2. Look inside the table of places
3. Put the places on the map
4. Colour the dots by type of place
5. Write the place names on the map
6. Explore the places
7. Measure distances
8. Make a layout and export your map

1

Open GeoLibre and Add India

First we need a map of India to put our places on. You already know how to do this!

1. Open GeoLibre in your web browser and click **Launch GeoLibre Web**.
2. Click **Add Data**, choose **Vector Layer**, and pick **india_boundary.geojson** from your Exercise 1 data folder.
3. In the **Style** panel, set the **Fill** to transparent and the **Stroke** to dark grey, width **2**, just like in Exercise 1.
4. Pick a plain, light basemap. Our coloured dots will be easier to see on a calm background.

TIP

Forgot a step? Flip back to Exercise 1. Repeating a skill is how real mapmakers get fast.

2

Look Inside the Table

Before we map anything, let's see what our data looks like.

1. Find **famous_places.csv** in the Exercise 2 data folder and open it in Notepad or Excel.
2. Each **row** is one place. Each **column** is one fact about it.
3. Find the `latitude` and `longitude` columns. These two numbers are what put each place on the map.
4. Look at the `type` column. Every place is a *Monument*, *Palace*, *Temple*, or *Nature* spot.

name	type	city	latitude	longitude
Taj Mahal	Monument	Agra	27.1751	78.0421
Hawa Mahal	Palace	Jaipur	26.9239	75.8267
Golden Temple	Temple	Amritsar	31.6200	74.8765
Kaziranga National Park	Nature	Golaghat	26.5775	93.1711

Four of the 16 rows in `famous_places.csv`. There's also a `state` and a `fun_fact` column.

FUN FACT

A CSV file is just plain text with commas between the values. CSV stands for **Comma-Separated Values**. Open it in Notepad and you'll see the commas!

3

Put the Places on the Map

Now let's turn that table into dots on the map.

1. Click **Add Data**. This time, under **Files**, choose **Delimited Text Layer**.
2. Pick **famous_places.csv**.
3. Tell GeoLibre which columns hold the coordinates: `longitude` for X (east-west) and `latitude` for Y (north-south).
4. Confirm. Sixteen dots pop up all over India!

GIS WORD OF THE DAY

Point data: places shown as single dots. In Exercise 1 you used shapes (polygons) for the country; here each place is so small on a map of India that one dot is enough.

4

Colour the Dots by Type

All the dots look the same right now. Let's give each type of place its own colour.

1. Click the *famous_places* layer in the Layers panel. The **Style** panel opens on the right.
2. Change **Style type** to **Categorized**.
3. Set **Attribute** to `type`.
4. Choose a **Colormap** with bright, different colours and click **Apply style type**.
5. Make the dots a little bigger so they're easy to spot, and give them a white outline.

GIS WORD OF THE DAY

Categorized style: sorting things into groups based on a column, and giving each group its own colour. Here there are four groups, one for each type.

5

Name the Places

Which dot is the Taj Mahal? Let's write the names on the map.

1. With *famous_places* selected, find the **Labels** settings in the Style panel.
2. Turn labels on and choose `name` as the label field.
3. Set the text colour to **black** and the size to about **12**.
4. Add a **white halo** about **1.5** wide, so the names are easy to read on top of the map.
5. Zoom in on Delhi. The Red Fort and Qutub Minar are so close that their names only separate when you zoom in.

GIS WORD OF THE DAY

Label: text placed on a map, taken straight from a column in the data. Change the name in the table, and the label changes too.

6

Explore the Places

Our map knows more than just names. Let's find the hidden facts.

1. Open the **Layer actions** for *famous_places* and choose **Open attribute table**. It's the same table as the CSV, now inside GeoLibre.
2. Click a row, for example *Hawa Mahal*. Its dot lights up on the map.
3. Scroll right to the `fun_fact` column and read it out loud!
4. Now **right-click** on the map near a place and choose **What's here?** A card pops up with information from Wikipedia.

FUN FACT

Hawa Mahal means "Palace of Winds". It has 953 small windows, so the royal ladies could watch the street below while staying cool and unseen.

Maps are great for measuring. Let's find out how far apart some of our places are.

1. Open **Controls** → **Measure** and pick **Distance**. Set the unit to kilometres.
2. Click **Start**. Click once on the **Taj Mahal**, then double-click on the **Gateway of India** to finish.
3. Read the distance. Write it down: _____ km
4. Click **Clear All**, then measure from the most northern place (the **Golden Temple**) to the most southern place (**Meenakshi Temple**): _____ km

GIS WORD OF THE DAY

Straight-line distance: the distance "as the crow flies", in a perfectly straight line. Roads twist and turn, so a car trip is always longer. *Check: you should get roughly 1,060 km and 2,430 km.*

Time to turn your map into something you can print and show off.

1. Open **Project** → **Print Layout...**
2. Type "**Famous Places of India**" as the title.
3. Turn on the **north arrow**, the **scale bar**, and the **legend**. The legend explains what each dot colour means.
4. Set the paper to **A4, Portrait**.
5. Export your map as a **PNG** image or a **PDF**.

GIS WORD OF THE DAY

Legend: the key to a map. It tells readers what each colour or symbol means.

Well Done, Explorer!

You turned a plain table into a map, explored it, and measured real distances across India. That's real GIS work!

What You Learned

Every place has **coordinates**: latitude and longitude

Turning a CSV table into **point data**

Colouring points with a **categorized style**

Writing names on a map with **labels**

Reading facts in the **attribute table**

Measuring **straight-line distance**

Try It Yourself

- **Add your own place.** Right-click on your school or home in GeoLibre and copy its coordinate. Add a new row to the CSV in Notepad, then load it again. Is your dot in the right place?
- Which two places on the map are closest together? Measure them to check.
- Switch the basemap to **Google Satellite** and zoom right in on the Taj Mahal. Can you see the white dome from space?
- Pick your favourite place on the map and find three more facts about it.